SB2 · STATIC-MEMBRANE

NAFEMS LE1 elliptic membrane (tangential edge stress at D)

In-plane membrane stress at a stress raiser — custom shell front-line check

PASS

error −2% · tol 3%

§2 TECHNICAL FEATURES TESTED

- **STRIX engine 4-node shell** — membrane with drilling rotations
- Production wall element, plane-stress membrane section
- Membrane placed in the vertical XZ plane, as production walls are
- Gauss-point membrane stress resultants
- h-refinement convergence study

§3 PROBLEM DESCRIPTION

A quarter of an elliptic annulus is loaded by a uniform outward normal tension on its outer elliptic edge B–C, with symmetry restraints on the two straight edges. The tangential edge stress σ_{yy} at the inner point D is a classic stress-raiser and the NAFEMS acceptance quantity. It is recovered as the maximum principal membrane stress at the Gauss point nearest D (at D the symmetry axis and the free inner edge make σ_{yy} a principal stress, so the value is frame-invariant). The membrane is modelled in the vertical XZ plane, as production walls are (membrane x → global X, membrane y → global Z); the plane problem and its answer are unchanged. First-order quadrilaterals approach the reference from below, so the element is exercised over a mesh-refinement sequence.

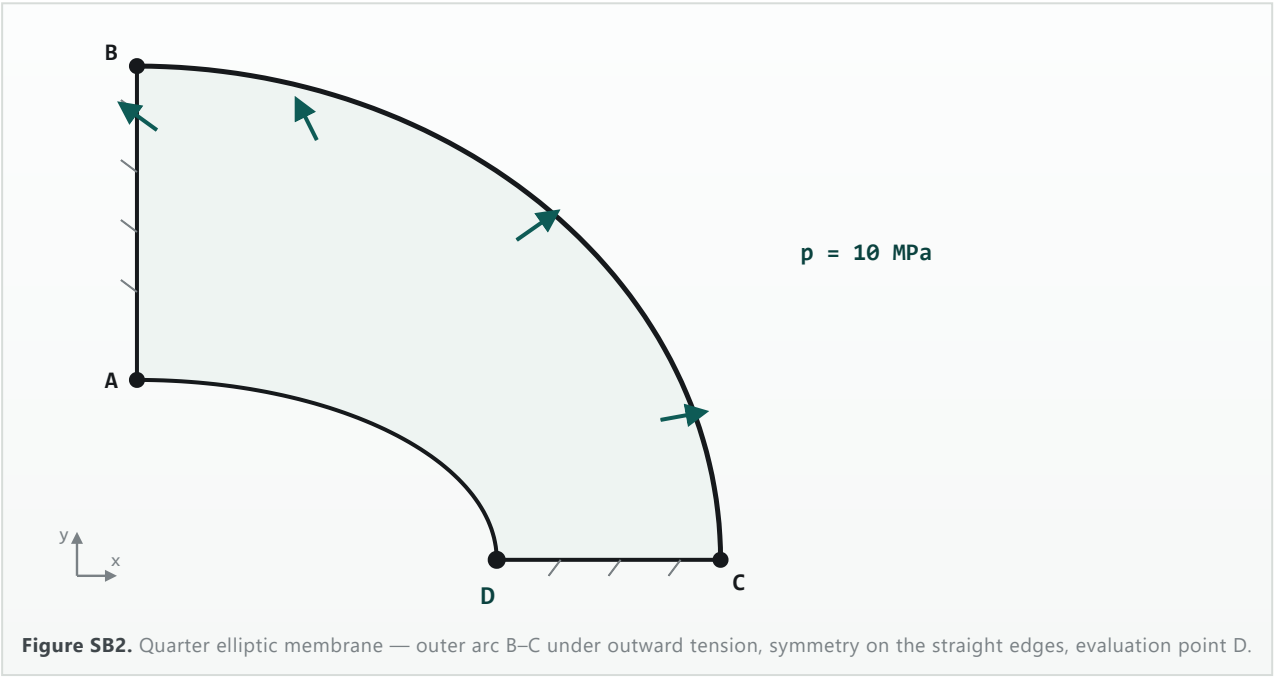


Figure SB2. Quarter elliptic membrane — outer arc B–C under outward tension, symmetry on the straight edges, evaluation point D.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MATERIAL		LOADING & BC	
Inner semi-axes	2000 × 1000 mm	Model	Elastic · plane stress	Outer arc B–C	10 MPa (tension)
Outer semi-axes	3250 × 2750 mm	E	210000 MPa	Symmetry	u_x=0 (AB), u_y=0 (CD) — membrane axes
Point D	2000, 0 mm	ν	0.3	Placement	vertical XZ plane (membrane y → global Z)
Thickness t	100 mm	Units	N · mm	Out-of-plane	global u_Y, r_X, r_Z fixed; r_Y free
				Analysis	Linear static

§5 REFERENCE SOLUTION

Tangential edge stress sigma_yy at D: sigma_yy(D) = max principal of (N11,N22,N12)/t at the Gauss point nearest D = 92.7 MPa

NAFEMS, The Standard NAFEMS Benchmarks (Linear Elastic), Benchmark LE1 — elliptic membrane; reference tangential edge stress sigma_yy = 92.7 MPa at point D.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Tangential edge stress sigma_yy at D (MPa)	mesh 96×48	90.8464	92.7	−2%	PASS

§7 CONVERGENCE

MESH	SIGMA_YY@D (MPA)	DIFF
8×4 cells	71.5038	−22.865%
16×8 cells	81.8126	−11.745%
24×12 cells	85.4283	−7.844%
32×16 cells	87.2414	−5.888%
48×24 cells	89.0488	−3.939%
64×32 cells	89.9552	−2.961%
96×48 cells	90.8464	−2%

The STRIX engine quadrilateral membrane converges to the NAFEMS reference `monotonically from below` as the mesh is refined, from −22.9% on the coarsest 8×4 mesh to −2.00% on the finest 96×48 mesh — consistent with first-order quadrilateral behaviour for this stress raiser. The trend confirms the membrane formulation converges to the correct continuum stress.

§8 CONCLUSION

PASS Across the refinement sequence the recovered $\sigma_{yy}(D)$ rises monotonically toward the NAFEMS reference of 92.7 MPa, and the finest mesh lies within tolerance. The STRIX engine wall shell is verified to converge to the reference membrane stress.

§9 REFERENCES & ANALYSIS FILES

1. NAFEMS — *The Standard NAFEMS Benchmarks (Linear Elastic Tests).* Benchmark LE1, elliptic membrane, $\sigma_{yy}(D)$=92.7 MPa.
2. STRIX headless harness: `generatePyForLoadCase → STRIX engine (static-single route) → wall_stresses.out`.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SB2/` `sha256 (pending publish)`